Monte Carlo analysis of local distribution of negative refractive index in nanosphere-doped liquid crystal metamaterial

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ABSTRACT

Khoo et al.[1] have introduced the concept of tuning of negative refractive index using nanosphere dispersed nematic liquid crystal (NDLC). Recently,[2,3] we have discussed anchoring forces as a new control parameter for negative-positive refraction index tuning in NDLC. In particular, we have calculated[3] the phase diagrams in variables electric field - anchoring force for real and imaginary parts of permittivity and permeability for NDLC using averaged (global) value of refractive index for the inhhomogeneous system. In current paper we analyze an influence of anchoring forces and of spatial modulation of electric field on local distribution of negative refraction index in NDLC. The study constitutes a generalization of homogenous isotropic dielectric layer approximation used in papers.[1] Inhomogeneous molecular order in planar NLC cells is modeled using Monte Carlo simulations with Lebwohl - Lasher effective hamiltonian and Rapini - Papoular term for anchoring forces.

Keywords: negative index of refraction, metamaterials, nematic liquid crystals, Monte Carlo

1. INTRODUCTION

Since ten years metamaterials have been in center of interest because of their potential unusual optic and photonic applications. Tunability of negative-zero-positive refractive index metamaterials is particularly desirable. Seminal paper of Khoo and co-workers[1] generalizes the approach of Wheeler at al.[4] by using nematic liquid crystal (NLC) as a medium in which the coated nanospheres are dispersed. The nanosphere–dispersed liquid crystal (NDLQ) model system displays negative real part of diffraction index at infrared frequencies. In paper[1] only uniform NLC order was studied. Successive applications of this approach were published in Refs.[5,6]

Inhomogeneous order in NLC host may be a direct cause of a non–uniform spatial distribution of negative refractive index, giving rise to tunable structure of inhomogeneous metamaterials. The simplest way to induce an inhomogeneous order in a planar NLC cell with anchoring forces is to apply either constant or modulated electric (magnetic) field.

Monte Carlo simulations enable a detailed "microscopic" study of inhomogeneous order in NLC systems. In particular, spatial distribution of refractive index can be calculated. The nature of orientational NLC order depends on amplitudes of two control parameters: anchoring force α and applied electric field E. The phase diagram in $\alpha - E$ variables calculated in[7] displays a new line of transitions caused by anchoring forces. Further extensions of the method, in particular an application to twisted NLC displays, were presented in papers.[2,8,9]

We have shown[2,3] that the amplitude of anchoring forces constitutes a new control parameter for tuning of negative–positive values of refraction index in NDLC.

The aim of this letter is to analyze the spatial inhomogeneity of refractive index in NDLC metamaterial for chosen values of incident light wavelength λ, anchoring force α and external electric field E with spatial modulation.

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2. THEORETICAL BACKGROUND

The effective refractive index of NDLC system was calculated using the approach introduced in papers.[1,4] Electromagnetic wave scattering from a single coated sphere was calculated using Mie theory;[10] the effective scattering from an ensemble of coated spheres dispersed in NLC matrix was analyzed in effective medium theory, using Maxwell Garnet mixing rule.[10–12] This approach relates an effective refractive index of NDLC metamaterial to the parameters of its constitutive components: coated nonmagnetic spheres, and host NLC system. The schematic shape of a single coated sphere is presented in Figure 1. Below, we present the basic formulas used for calculations.

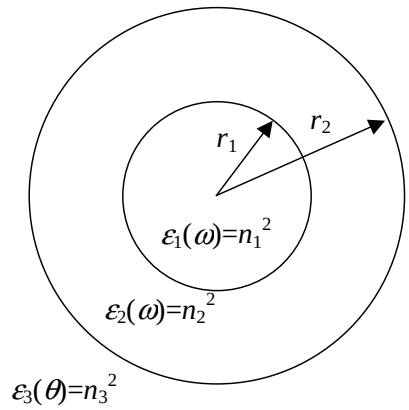

Figure 1. Parameters of the coated sphere (see text for details).

The core with radius r_1 is made from polaritonic material (such as $LiTaO_3$) with relative permittivity

$$\epsilon_1(\omega) = \epsilon_1(\infty)\left(1 + \frac{\omega_T^2 - \omega_L^2}{\omega_T^2 - \omega^2 - i\omega\gamma_1}\right). \tag{1}$$

Here $\epsilon_1(\infty)$ denotes the high-frequency dielectric constant, γ_1 – damping coefficient, ω_T and ω_L – transverse and longitudinal phonon frequencies, respectively. Required relative permittivity is of the order of magnitude 10^2.[4] Bulk cavity resonance in the core is responsible for negative effective permeability of the assembly of spheres at particular electromagnetic wavelengths.

The relative permittivity of the semiconductor shell with outer radius r_2 is described by Drude model:

$$\epsilon_2(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\omega\gamma_2}, \tag{2}$$

where ω_p denotes plasma frequency and γ_2 – the damping coefficient. The shell is responsible for negative effective permittivity, with ω_p close to frequency for which negative index will be detected. Its value is chosen to provide both negative effective permittivity and permeability in some interval of wavelengths.[4]

Permittivity of the liquid crystal host for extraordinary polarized wave depends on angle θ between directions of incident light wave vector and director of NLC:[13]

$$\epsilon_3(\theta) = \frac{\epsilon_\parallel \epsilon_\perp}{\epsilon_\parallel \cos^2(\theta) + \epsilon_\perp \sin^2(\theta)}, \tag{3}$$

where $\epsilon_\parallel$ and $\epsilon_\perp$ are permittivities for parallel and perpendicular orientations of director with respect to incident wave vector $\vec{k}_0$ in vacuum. In pure NLC host (without spheres) $k_3 = \sqrt{\epsilon_3}\, k_0$.

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Electromagnetic wave Mie scattering was calculated in dipolar (electric and magnetic) approximations, with amplitudes of scattered fields proportional to coefficients a_1 and b_1, respectively. The detailed expressions can be found, e.g., in.[10] Those coefficients enter the expressions for effective permittivity and permeability of NDLC, calculated in effective medium theory:[1]

$$\epsilon_{eff} = \epsilon_3 \left(\frac{k_3^3 + 4\pi i N a_1}{k_3^3 - 2\pi i N a_1} \right), \tag{4}$$

$$\mu_{eff} = \frac{k_3^3 + 4\pi i N b_1}{k_3^3 - 2\pi i N b_1}. \tag{5}$$

Those formulas are valid only for small number density $N = 3f/4\pi r_2^3$ of spheres (f is the filling factor) and in long-wavelength limit $2\pi/k_3 \gg r_2$. The effective index of refraction was calculated as

$$n_{eff} = \pm\sqrt{\epsilon_{eff}\ \mu_{eff}}, \tag{6}$$

where we have used the root with a positive imaginary part. Khoo et al. have shown[1] that for some values of the parameters of the model there is a range of frequencies ω and angles θ for which

$$n'_{eff}(\omega, \theta) \equiv \Re\big(n_{eff}(\omega, \theta)\big) < 0. \tag{7}$$

3. MONTE CARLO SIMULATION

In our model the NLC molecules, represented here by stiff rods, occupy sites of a $n_x \times n_y \times n_z$ rectangular lattice. Translational motions of the molecules are prohibited. An orientation of a long axis of a NLC molecule in an external coordinate system is described by a pair of angles ϕ, θ. Angle ϕ (twist) describes an azimuth and angle θ (tilt) is a polar angle calculated from z–axis, along which the electromagnetic wave propagates. Along the x and y-direction the periodic boundary conditions are applied, while the molecules in the lowest and the uppermost planes ($k = 1$ and $k = n_z$) are subject to properly chosen anchoring forces due to their interaction with a substrate.

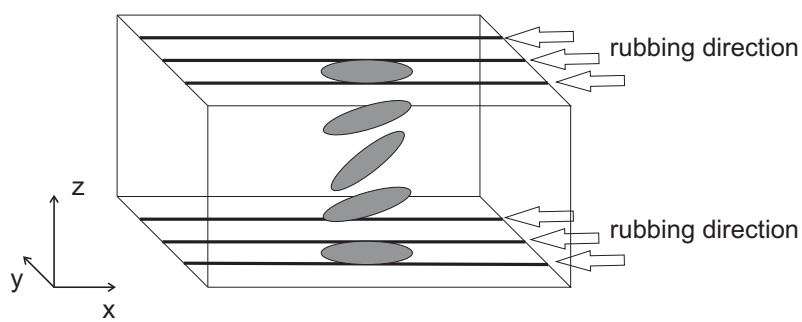

Figure 2. Geometry of the model.

Interactions are described by the Lebwohl–Lasher effective hamiltonian[14] with Rapini–Papoular term:[15]

$$H = -\xi \sum_{<\vec{r},\vec{r}\,'>} P_2(\cos\beta(\vec{r},\, \vec{r}\,')) - \sum_{\vec{r}} E^2(\vec{r}))\, P_2(\cos\beta(\vec{r})) + \sum_{\vec{r}_W} \alpha(\vec{r}_W)\, \sin^2\gamma(\vec{r}_W), \tag{8}$$

where P_2 denotes the second-order Legendre polynomial. Parameter ξ ($\xi = 25$ in 3D) is the coupling constant for NLC–NLC orientational interaction, $E(\vec{r})$ is external electric field, $\alpha(\vec{r}_W)$ – the amplitude of anchoring forces acting on molecules located at the walls. $\beta(\vec{r},\, \vec{r}\,')$ denotes the relative angle between two molecules located at points $\vec{r}$, $\vec{r}\,'$, $\beta(\vec{r})$ – the angle a molecule located at point $\vec{r}$ makes with electric field direction. $\gamma(\vec{r}_W)$ is the angle between long axis of molecule at the wall and a fixed rubbing direction. The rubbing direction is parallel to x

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axis (Fig. 2). Each molecule placed at the site is surrounded by the six nearest neighbors and interactions with those neighbors are taken into account.

The equilibrium configurations of NLC molecules on a lattice were sampled using Metropolis Monte Carlo simulations.[16,17]The system was equilibrated during 3×10^4 MC steps (MCS) and then simulated during 2×10^5 MCS. From 2×10^5 equilibrium configurations generated during simulation 2×10^2 MCS were used (each 10^3 MCS) to calculation of averaged values of chosen physical properties.

We have simulated the system for spatially modulated electric field. The field consists of constant part E_{DC} and sinusoidal modulated part with amplitude A (Fig. 3):

$$E(x) = E_{DC} + A\sin(qx). \tag{9}$$

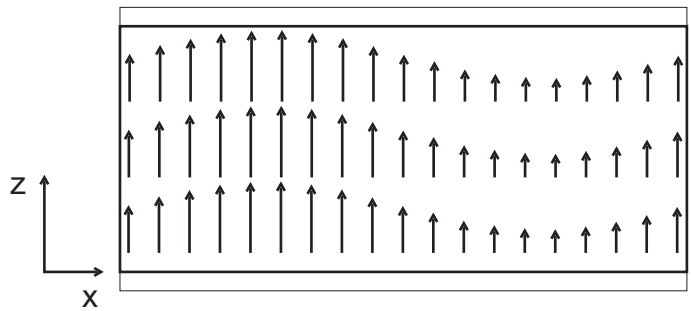

Figure 3. Spatially modulated electric field in NDLC cell.

This modulation may be a result of sinusoidally modulated light intensity in two-wave mixing experiment.[18] The supposition that the sinusoidal light intensity pattern produces a respective modulation of electric field has already been proved in many LC systems containing either photoconducting polymeric or semiconductor electrodes[19] or photoconducting LC mixtures.[20–22]

4. CALCULATION OF 2D MAPS OF DISTRIBUTION OF REFRACTIVE INDEX

For each of 2×10^2 equilibrium configurations (LC molecules orientations) local value of real part of refractive index was calculated. Fig. 4 shows the functional dependence of real part of the (local) refractive index n'_{eff} on NLC particle orientation (polar angle θ), calculated for parameters used by Khoo et al.:[1] $r_1 = 0.13\ \mu\mathrm{m}, \mathrm{r}_2 = 0.143\ \mu\mathrm{m}, \mathrm{f} = 0.12, \epsilon(\infty) = 17, \omega_\mathrm{T}/2\pi = 240$ THz, $\omega_L/2\pi = 570$ THz, $\gamma_1/2\pi = 2.5$ THz, $\omega_\mathrm{p}/2\pi = 134$ THz and $\gamma_2 = \omega_p/60$.

Fig. 5 shows a way of calculation of 2D maps of distribution of refractive index in $x - z$ plane: orientations of LC molecules for chosen cross-section in $x - z$ plane (a); the map of local value of real part of refractive index calculated for these molecules (b); two–dimensional maps of spatial distribution of $n'_{eff}(x, y, z)$ averaged over y axis (c); final 2D map after averaging over 2×10^2 equilibrium configurations (d).

5. RESULTS FOR SPATIALLY MODULATED ELECTRIC FIELD

Spatial modulation of applied external electric field and spatial modulation of the strength of anchoring forces has a strong impact on resulting inhomogeneous order in planar NLC cell.[2,7–9,23] It determines the spatial distribution of refractive index which, for some values of parameters, can assume negative values. Negative refraction index requires a fine tuning of the parameters of NDLC.[1]

For linearly polarized light with $\lambda = 2.830\ \mu$m (see Fig. 4) incident as an extraordinary wave on NLC host, NDLC may form a metamaterial with $n'_{eff} < 0$ if electric field (E_{DC}) is sufficiently large. For electric field below the Freedeericksz threshold metamaterial is not observed (Fig. 6 (a)). For E_{DC} close to Freedeericksz threshold

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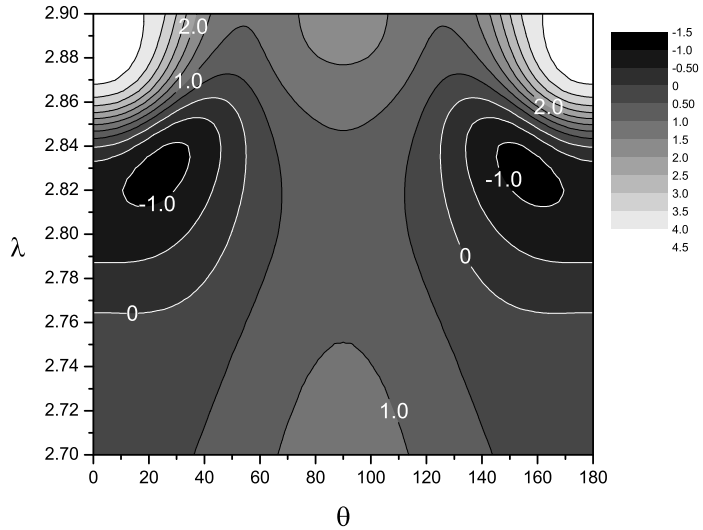

Figure 4. Real part of the local refractive index vs. NLC particle orientation (θ angle).

regions with $n'_{eff} < 0$ are observed (Fig. 6 (b)) when $E \approx E_{DC} + A$. In Fig. 6 (a) and (b) maps for strong anchoring forces ($\alpha = 50$) are presented. For this regime the molecules adjacent to walls are parallel to them resulting in $n'_{eff} > 0$. On the contrary for weak anchoring and the same value of electric field as for Fig. 6 (b) ($E_{DC} = 0.8$) we get a metamaterial for almost whole NDLC cell. Regions marked with striped pattern have negative values of refractive index.

6. CONCLUSIONS

We have applied lattice Monte Carlo simulations to the nanosphere dispersed liquid crystal model proposed recently by Khoo et al.[1] The studied system was a planar cell in a spatially modulated electric field. The inhomogeneous NLC order gave rise to a spatially inhomogeneous distribution of refractive index.

We have shown that the NDLC may consist of regions of normal material with inclusions of metamaterial. Even a small variation of incident light wavelength, electric field strength and anchoring forces may lead to very different inhomogeneous spatial organization of metamaterial. Demonstrated effect of tuning of spatial organization of NDLC can be of interest for photonic applications.

We have analyzed the 2D maps of spatial distribution of refractive index only. However, the transmission and absorption coefficients for such a system can also be calculated. For strictly layered systems those topics were reported in papers.[24–26]

Similar studies for constant external field and strong anchoring forces were presented in paper.[3]

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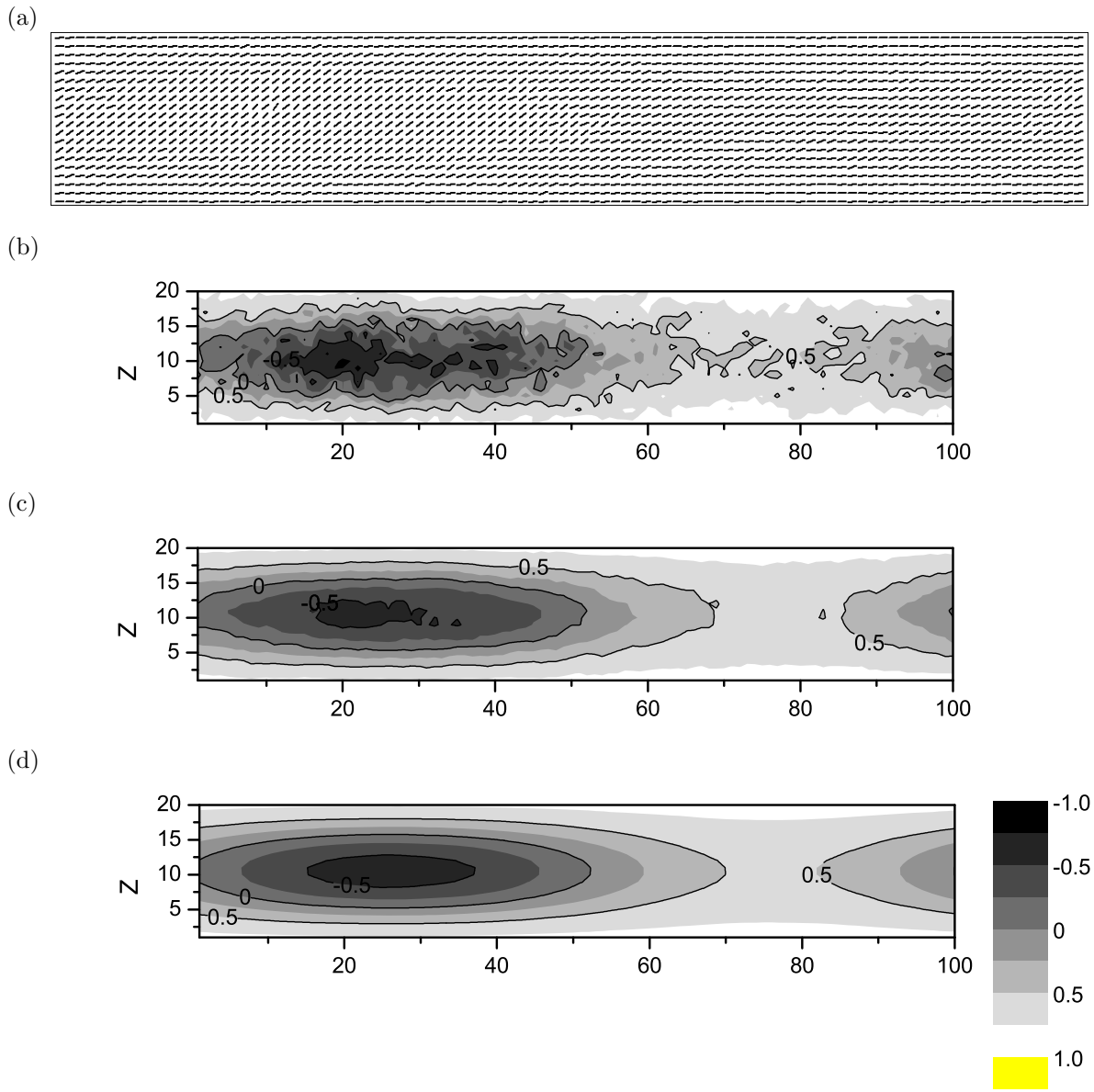

Figure 5. The way of calculation of 2D maps of spatially distribution of refractive index for $A = 0.08$, $\lambda = 2830$ nm, $\alpha = 30$ and $E_{DC} = 0.8$.

REFERENCES

1. I.C. Khoo, D.H. Werner, X. Liang, and A. Diaz, Opt. Lett. **31**, 2592 (2006).
2. G. Pawlik, A.C. Mitus, and F. Kajzar, Proc. SPIE **6654**, 66540G (2007).
3. W. Walasik, M. Jarema, G. Pawlik, R. Orlik, A. C. Mitus, and F. Kajzar, Proc. SPIE **7213**, 72130A (2009).
4. M.S. Wheeler, J.S. Aitchison, and M. Mojahedi, Phys. Rev. B **73**, 045105 (2006).
5. I.C. Khoo, A. Diaz, S. Kubo, J. Liou, M. Stinger, T. Mallouk, and J.H. Park, Mol. Cryst. Liq. Cryst. **485**, 186 (2008).
6. D.H. Werner, D.H. Kwon, I.C. Khoo, A.V. Kildishev, and V.M. Shalaev, Opt. Expr. **15**, 3342 (2007).
7. K. Komorowska, G. Pawlik, A.C. Mitus, and A. Miniewicz, J. Appl. Phys. **90** (2001) 1836-1842.
8. G. Pawlik, A.C. Mitus, and F. Kajzar, Proc. SPIE **5990**, 59900F (2005).

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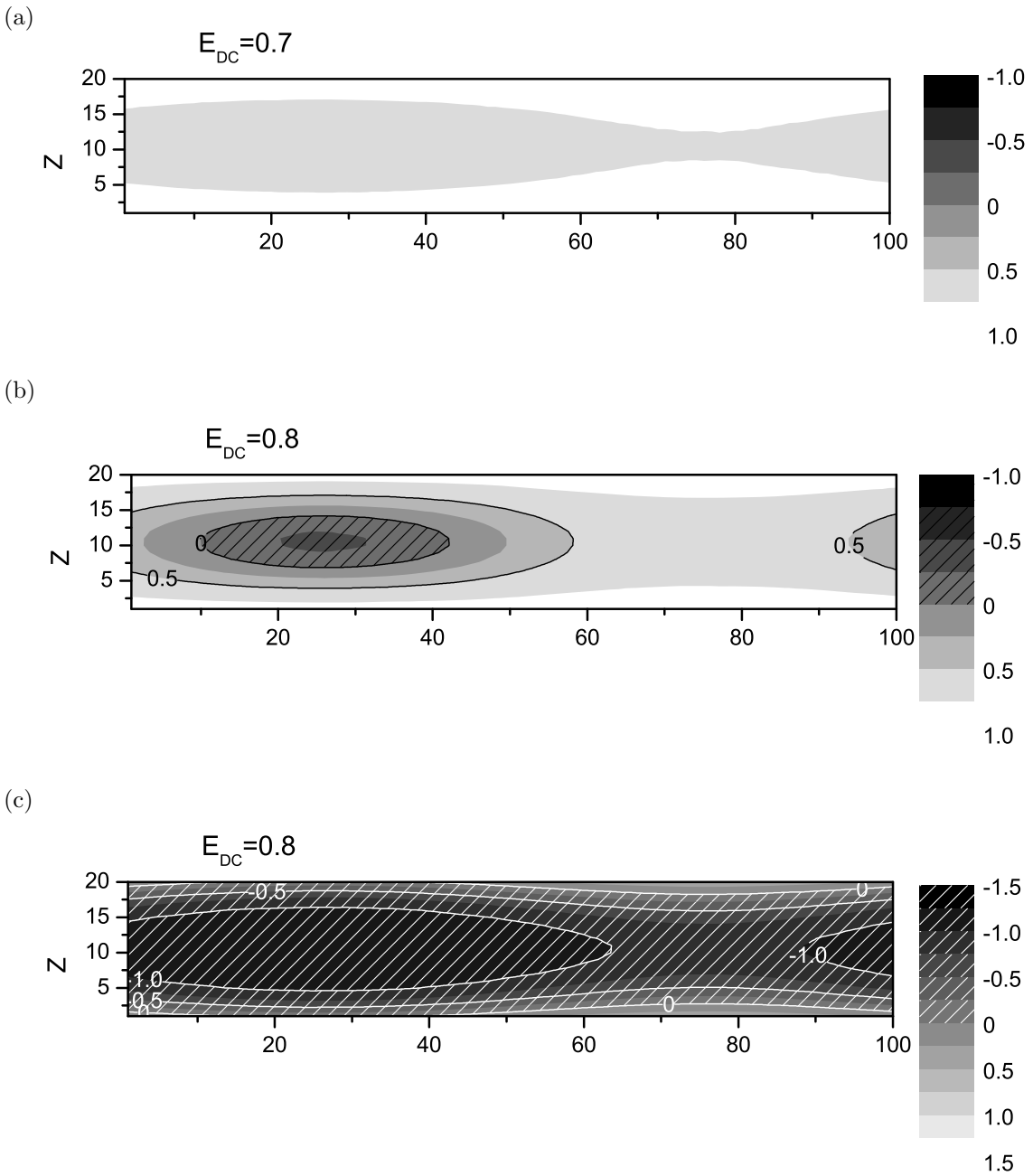

Figure 6. 2D maps of local real part of refractive index for $A = 0.08$, $\lambda = 2830$ nm and: $E_{DC} = 0.7$, $\alpha = 50$ (a); $E_{DC} = 0.8$, $\alpha = 50$ (b); $E_{DC} = 0.8$, $\alpha = 10$ (c). Regions marked with striped pattern have negative values of refractive index.

9. G. Pawlik, A.C. Mitus, and F. Kajzar, Proc. SPIE **6332**, 63320B (2006).
10. C.F. Bohren and D.R. Huffman, *Absorption and Scattering of Light by Small Particles*, Wiley, 1998.
11. J.C. Maxwell Garnett, Philos. Trans. R. Soc. Ser. A **203**, 385 (1904).
12. C.A. Guérin, P. Mallet, and A. Sentenac, J. Opt. Soc. Am. A **23**, 349 (2005).
13. I.C. Khoo, *Liquid crystals*, Wiley, 2007.
14. P.A. Lebwohl and G. Lasher, Phys. Rev. A **6**, 426 (1972).
15. A. Rapini and M. Papoular, J. Phys. (France) **30**, 54 (1969).

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16. P. Allen and D. Tildesley, *Computer Simulation of Liquids*, Clarendon Press, Oxford, 1987.
17. D. Landau and K. Binder, *A Guide to Monte Carlo Simulations in Statistical Physics*, Cambridge University Press, 2000.
18. A. Miniewicz, S. Bartkiewicz, W. Turalski, and A. Januszko, in *Electrical and Related Properties of Organic Solids*, edited by R. W. Munn, A. Miniewicz, and B. Kuchta NATO ASI Series 3/24 (Kluwer Academic, Dordrecht, 1997), pp. 323-337.
19. A. Miniewicz, S. Bartkiewicz, and F. Kajzar, Nonlinear Opt. **19**, 157 (1998).
20. I. Jánossy, A. D. Lloyd, and B. S. Wherrett, Mol. Cryst. Liq. Cryst. **179**, 1 (1990).
21. S. Bartkiewicz, A. Miniewicz, A. Januszko, and J. Parka, Pure Appl. Opt. **5**, 799 (1996).
22. I.-C. Khoo, IEEE J. Quantum Electron. **32**, 525 (1996).
23. G. Pawlik, A.C. Mitus, and A. Miniewicz, Opt. Commun. **182** (2000) 249-254.
24. R. Aylo, P.P. Banerjee, and G. Nehmetallah, Proc. SPIE **7029**, 702917 (2008).
25. T. Pan, C. Tang, L. Gao, and Z. Li, Phys. Lett. A **337**, 473 (2005).
26. Y. Dong and X. Zhang, Phys. Lett. A **359**, 542 (2006).

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